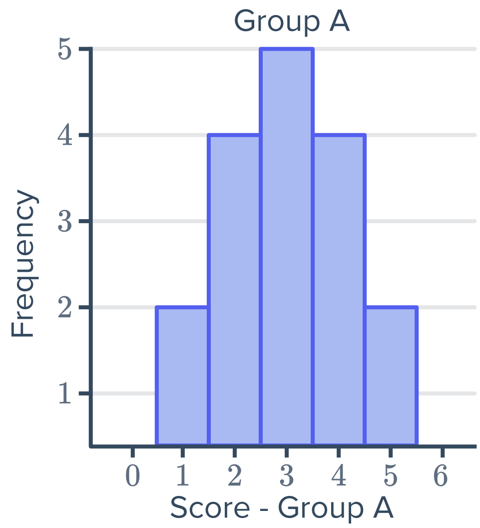
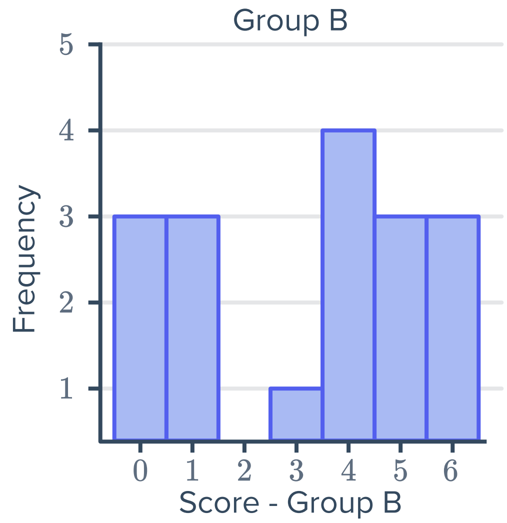
Compare data sets

1 The following histograms show the season results of two football groups, Group A and Group B, and the number of games (frequency) in which they scored a certain number of goals:

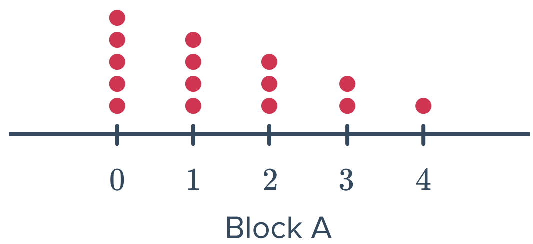


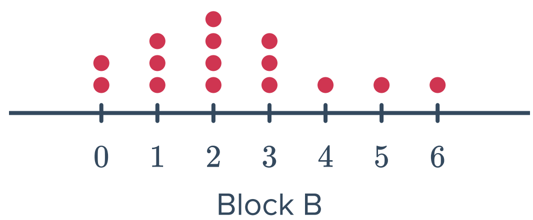


- **a** Find the mode for Group A.
- **b** Find the mode for Group B.
- **c** Find the range for Group A.
- **d** Find the range for Group B.
- **e** Which group scored the least total number of goals during the season?
- **f** Which group has the most varied results?

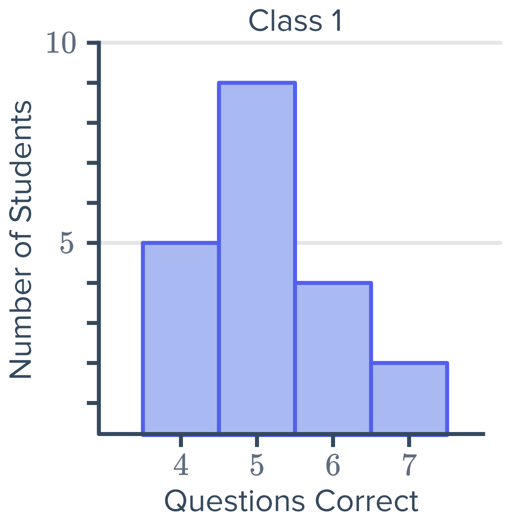
2 The residents of two blocks of townhouses were asked the number of pets they own. The frequency of various responses are presented in the dot plots.

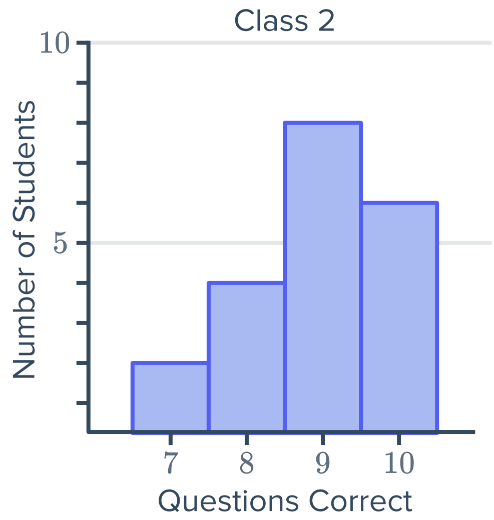
- **a** In which block is pet ownership less in general?
- **b** In which block do most households have zero or one pet?
- **c** In which block is there a greater range in the number of pets?
- **d** In which block is pet ownership skewed positively?
- **e** How many pets are there in total in block B?





3 Two Science classes, each with 20 studen[redacted]re given a 10 question True/False test. The results for each class are shown below:





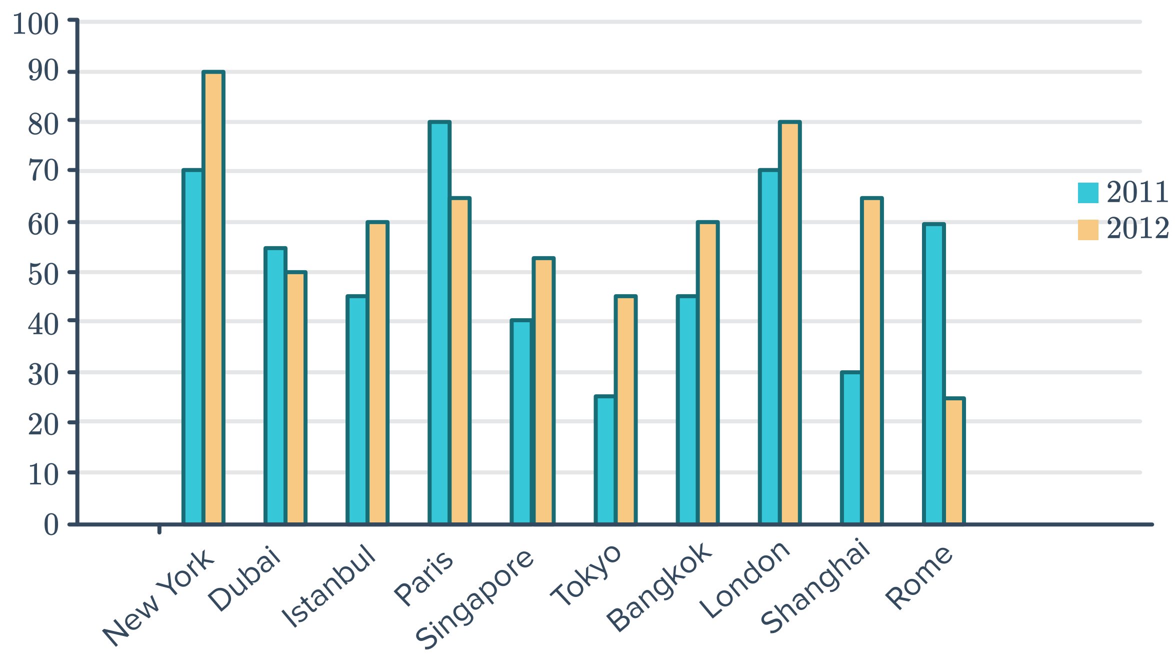
a Complete the following table:

	Mean	Range
Class 1		
Class 2		

b Which class was more likely to have studied effectively for their test? Explain your answer.

4 The median house price in the suburb of Humbleton is $\$950\,000$ with a mean price of $\$1\,000\,000$ and the median house price in the suburb of Brockway is $\$950\,000$ with a mean price of $\$880\,000$.
Which suburb is more likely to have very expensive houses? Explain your answer.

5 The following bar graph shows the change[redacted]ourism rates in different cities during 2011 and 2012:



a Which city had the greatest percentage of tourism in 2011?

b Which city had the least percentage of tourism in 2012?

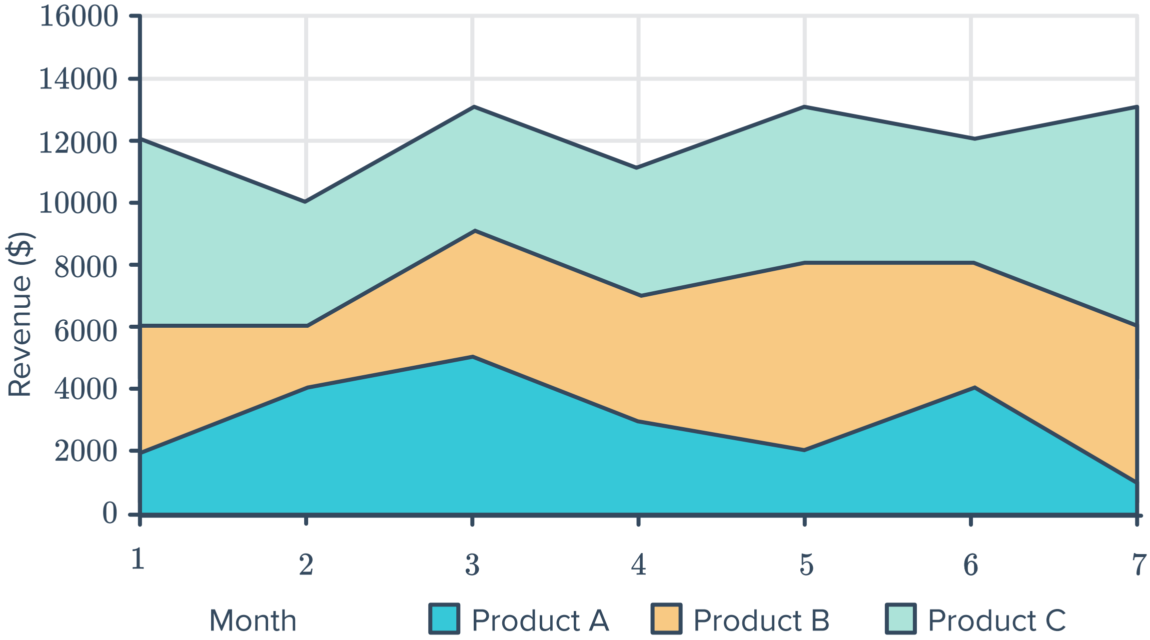
c Which city had the greatest percentage tourism in a single year?

d Which city/cities had the least percentage tourism in a single year?

e How much greater is Paris's percentage of tourism in 2011 than that of London in 2012?

f Paris' maximum percentage of tourism over the two years is greater than Istanbul's maximum by how many percent?

6 The revenue generated from a company's [redacted] major products is shown in the following stacked area chart:



a Find the revenue generated by Product C in the first month.

b Find the maximum revenue generated by Product A.

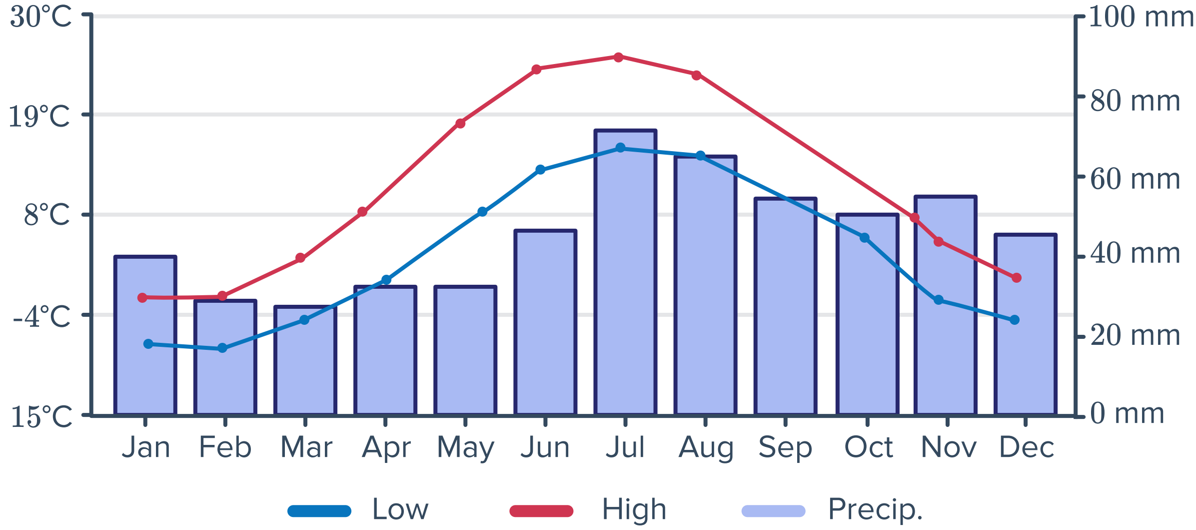
c Determine the difference between the revenue generated in month 4 and the revenue generated in month 5 by Product B.

d Calculate the total revenue generated by Product C during the 7 months.

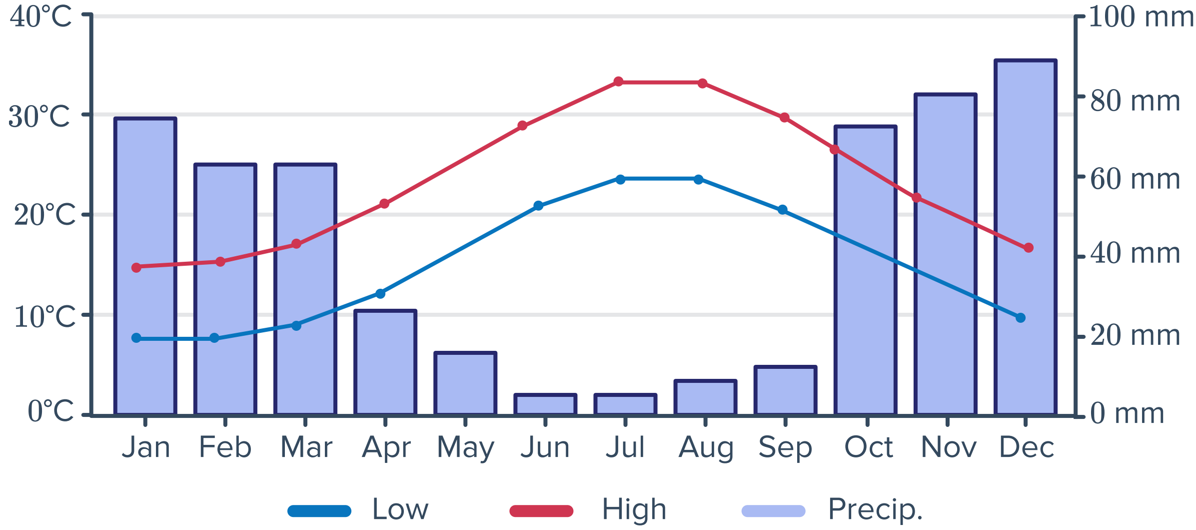
e By how much did the revenue generated by Product C exceed that of Product A in the 7th month?

7 Consider the following graphs:

Stockholm



Athens



a Which axis measures the precipitation (rainfall), vertical or horizontal axis?

b Which axis measures the temperature?

c Which city has greater average temperature?

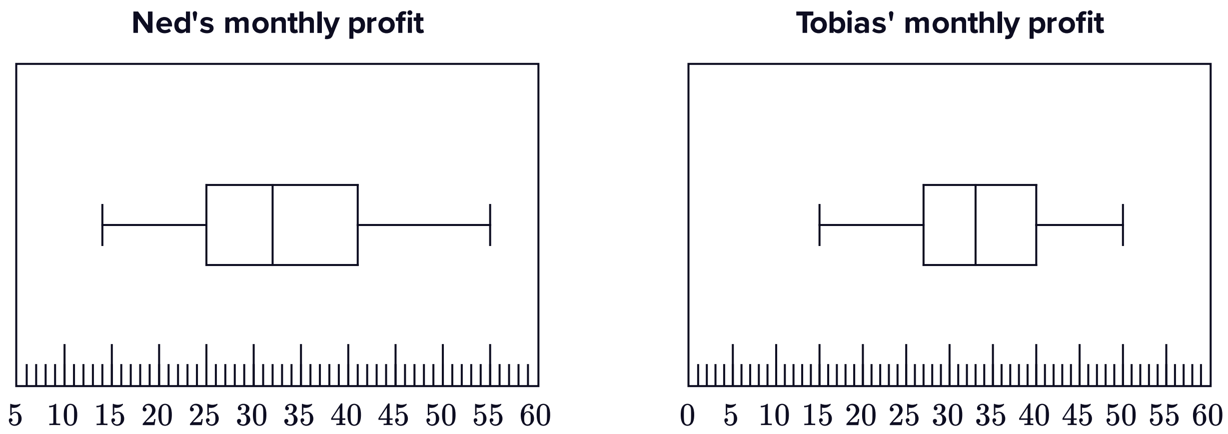
d Which city has the greater average rainfall?

e Determine whether the following statements are true or false:

i The greater the average temperature in a city, the greater the average rainfall.

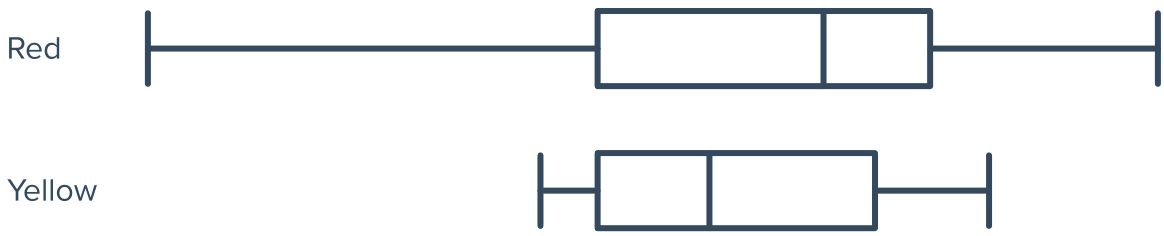
ii The average minimum temperature rises with the average maximum temperature.

8 The box plots show the monthly profits (in thousands of dollars) of two derivatives traders over a year:



- **a** Which trader made a greater median monthly profit?
- **b** Which trader's profits had a greater interquartile range?
- **c** Which trader's profits had a greater range?
- **d** How much more did Ned make in his most profitable month than Tobias in his most profitable month?

9 A builder can choose between two different types of brick that are coloured red or yellow. The box plots below illustrate the results of tests on the strength of the bricks:



- **a** Explain why a builder may prefer to use red bricks.
- **b** Explain why a builder may prefer to use yellow bricks.

10 The heights (in meters) of the boys and gi[redacted] class of 30 students were recorded. The results are given in the table below:

a Find the five number summary of the heights of boys in the class.

b Find the five number summary of the heights of girls in the class.

c Draw a parallel box plot for this data.

Boys	Girls
1.65	1.55
1.66	1.56
1.67	1.57
1.68	1.58
1.63	1.53
1.62	1.52
1.61	1.51
1.60	1.50
1.75	1.69
1.76	1.70
1.77	1.71
1.78	1.72
1.73	1.67
1.72	1.66
1.71	1.65

11 Two friends, Rhonda and Michael, have been growing sunflowers. They have each measured the heights of their sunflowers to the nearest centimeter. The data from these measurements is shown below:

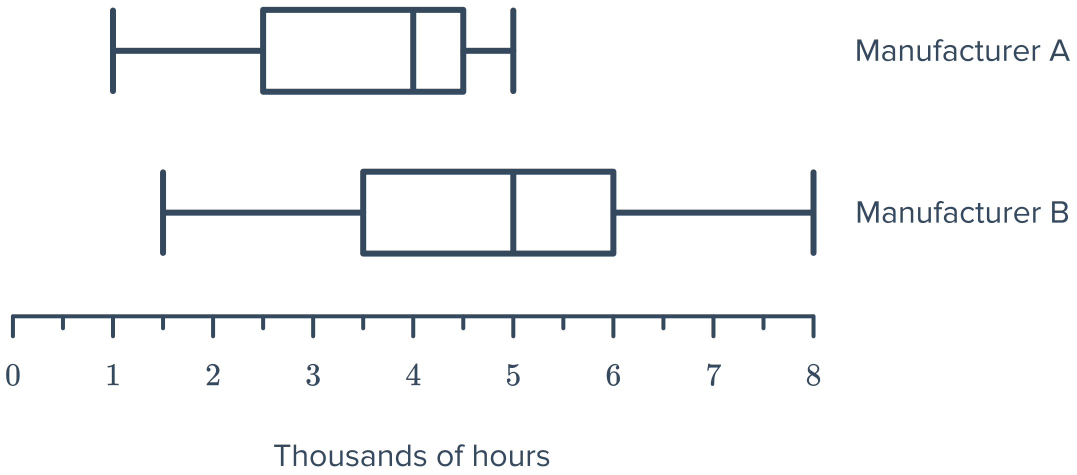
- Rhonda: $3, 7, 7, 11, 16, 21, 24, 27, 28, 31, 41, 46, 49$
- Michael: $13, 18, 19, 23, 28, 35, 39, 46$

a Find the five number summary for the heights of Rhonda's sunflowers.

b Find the five number summary for the heights of Michael's sunflowers.

c Draw parallel box plots for this data.

12 The two box plots below show the data co[illegible]ed by the manufacturers on the life-span of light bulbs, measured in thousands of hours:



a Copy and complete the following table for the two sets of data. Write each answer in terms of number of hours.

	Manufacturer A	Manufacturer B
Median		
Lower Quartile		
Upper Quartile		
Range		
Interquartile Range		

b Determine which manufacturer produces light bulbs with the longer lifespan?

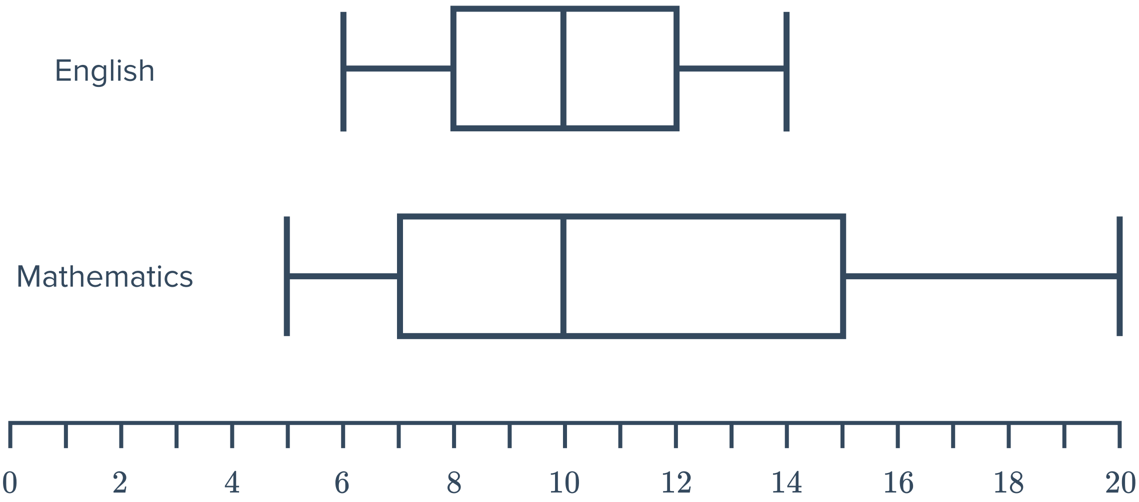
13 The heights (in meters) of the boys and girls in a class of 30 students were recorded. The results are given below:
Boy's heights: $1.65, 1.66, 1.67, 1.68, 1.63, 1.62, 1.61, 1.60, 1.75, 1.76, 1.77, 1.78, 1.73, 1.72, 1.71$
Girl's heights: $1.55, 1.56, 1.57, 1.58, 1.53, 1.52, 1.51, 1.50, 1.69, 1.70, 1.71, 1.72, 1.67, 1.66, 1.65$

a Find the five number summary for the heights of boys in the class.

b Find the five number summary for the heights of girls in the class.

c Draw a parallel box plots for this data.

14 A class took an English test and a Mathem[redacted]test. Both tests had a maximum possible score of 25. The results are illustrated below:



a Copy and complete the following table using the two box plots:

	English	Mathematics
Median		
Lower Quartile		
Upper Quartile		
Range		
Interquartile Range		

b In which test did the class tend to score better scores?

15 The test scores of 12 students in English and Music are listed below:

- English: $55, 57, 63, 69, 71, 74, 77, 81, 84, 88, 91, 98$
- Music: $55, 61, 66, 69, 72, 74, 76, 81, 84, 86, 89, 93$

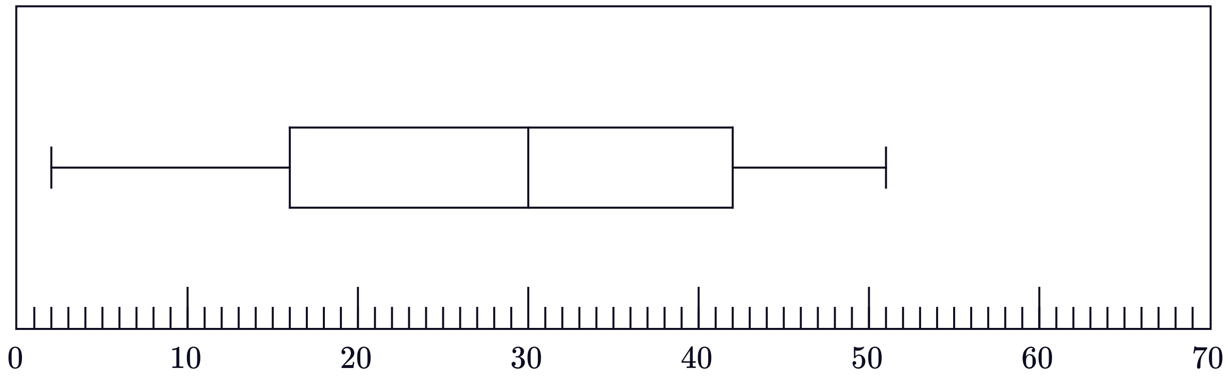
a Find the five number summary for the test scores for English.

b Find the five number summary for the test scores for Music.

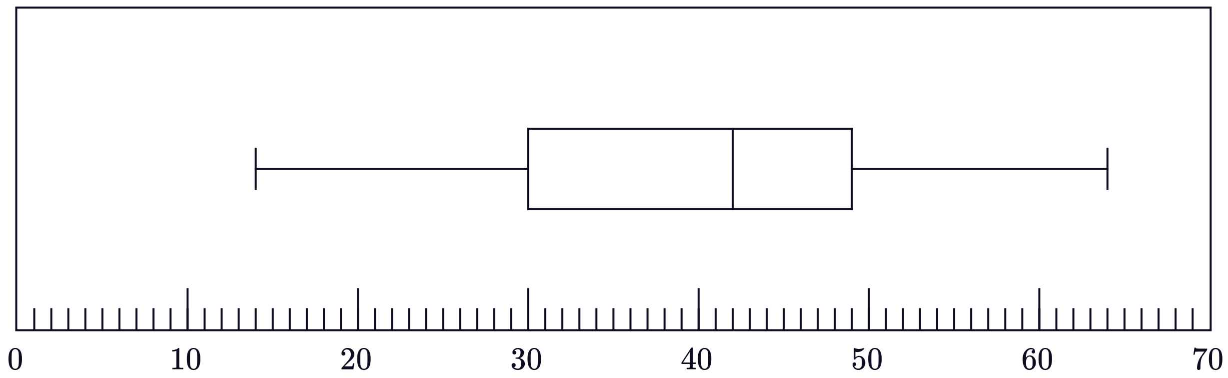
c Draw parallel box plots for this data.

16 The box plots below represent the daily [redacted]ade by Carl and Angelina over the course of one month:

Angelina's sales



Carl's sales



a Calculate the range for Angelina's sales.

b Calculate the range for Carl’s sales.

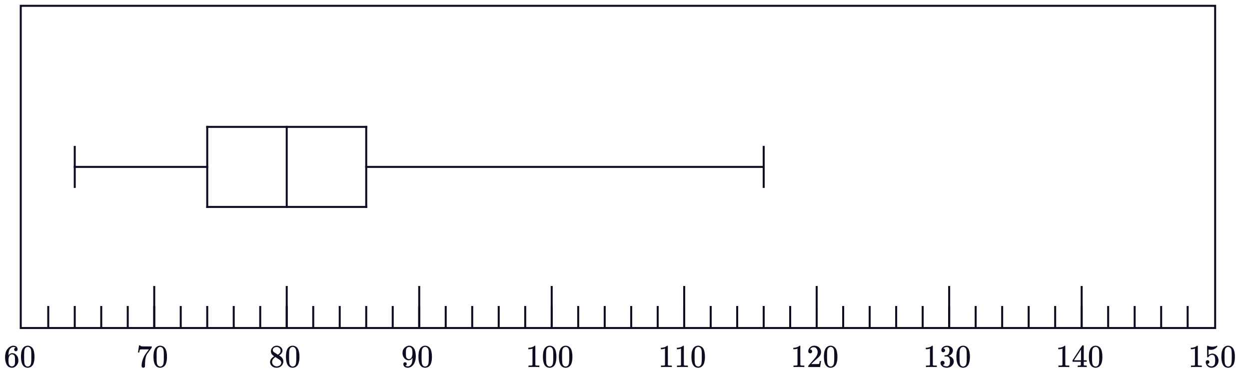
c By how much did Carl’s median sales exceed Angelina's?

d Considering the middle 50% of sales, which salesperson had the more consistent number of sales?

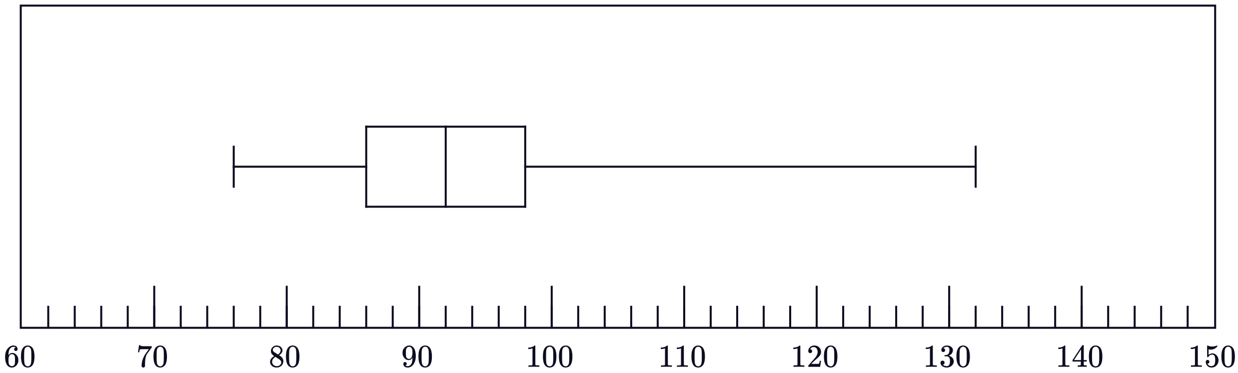
e Which salesperson had a more successful sales month?

17 At every training session of the season, a [redacted] measured her pulse rate before a sprint and after a sprint. The before and after rates, measured in beats per minute (bpm), recorded throughout the season are presented in the parallel box plots below:

Pulse rate before (bpm)

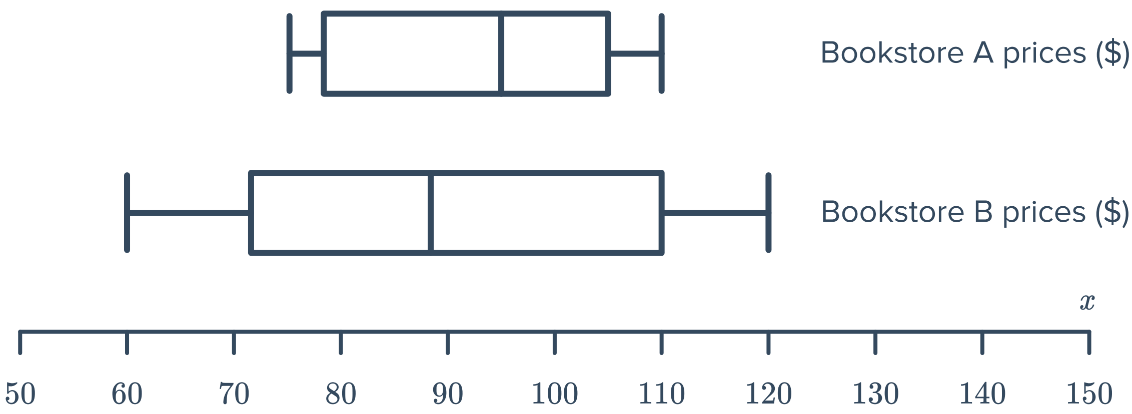


Pulse rate after (bpm)



- **a** By how much did her median pulse rate increase during the sprint?
- **b** Determine the interquartile range of her pulse rate before the sprint.
- **c** Determine the interquartile range of her pulse rate after the sprint.
- **d** Determine the range of her pulse rate before the sprint.
- **e** Determine the range of her pulse rate after the sprint.
- **f** Are her pulse rate readings more consistent before or after the sprint?
- **g** In the last session of the season, the cyclist recorded her greatest pulse rate of the season both before and after the sprint. By how much did her pulse rate increase during this particular training session?

18 Two bookstores recorded the selling price their books. The results are presented in the parallel box plots:



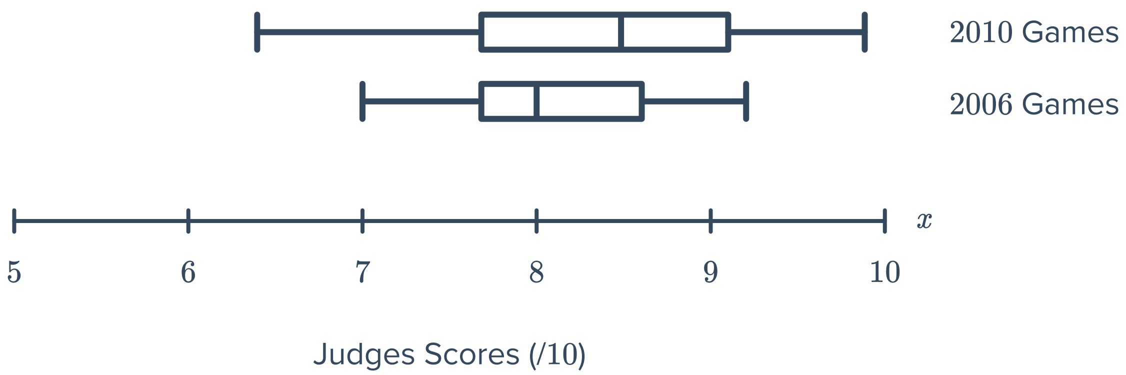
a Which bookstore had the more consistent prices?

b Comparing the most expensive books in each store, how much more expensive is the one in store B?

c True or False: 25% of the books in Bookstore B are at least as expensive than the most expensive book in Bookstore A.

d True or False: 25% of the books in Bookstore B are cheaper than the cheapest book in Bookstore A.

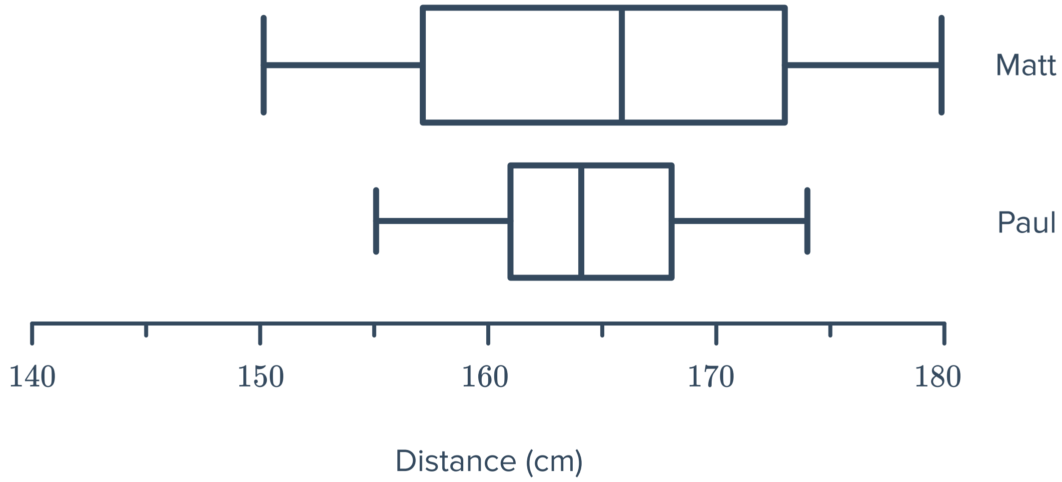
19 Eileen competed in the high beam gymnastics event at both the 2006 and 2010 Olympics. The judges' scores in both years are presented in the parallel box plots:



a In which year did the judges score her greater overall?

b In which year did the judges score Eileen most consistently?

20 The parallel box plots shows the distances in centimeters, jumped by two high jumpers:



a Who had a greater median jump?

b Who made the greatest jump?

c Who made the least jump?

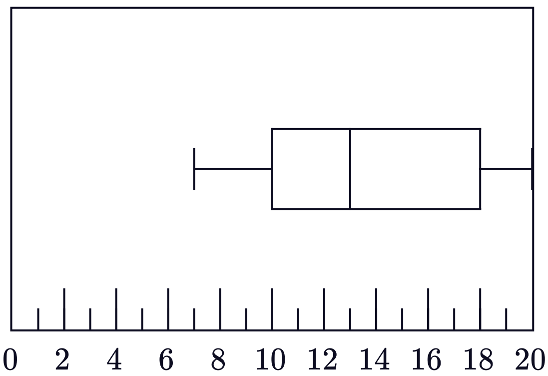
d Who is the most consistent?

21 A mathematics test is given to two classes. The scores out of 20 received by students in each class are represented in the box plots below:

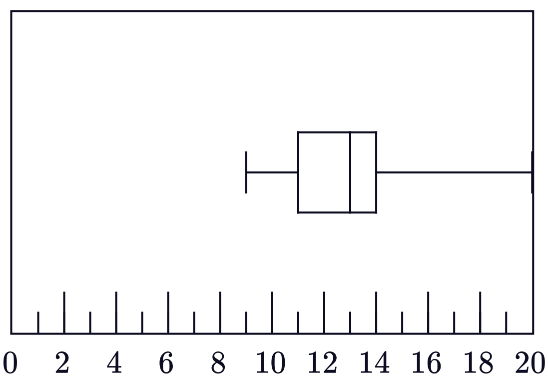
a Complete the following table:

	Class 9P	Class 9Q
Median		
Lower quartile		
Upper quartile		
Range		
Interquartile range		

Class 9P

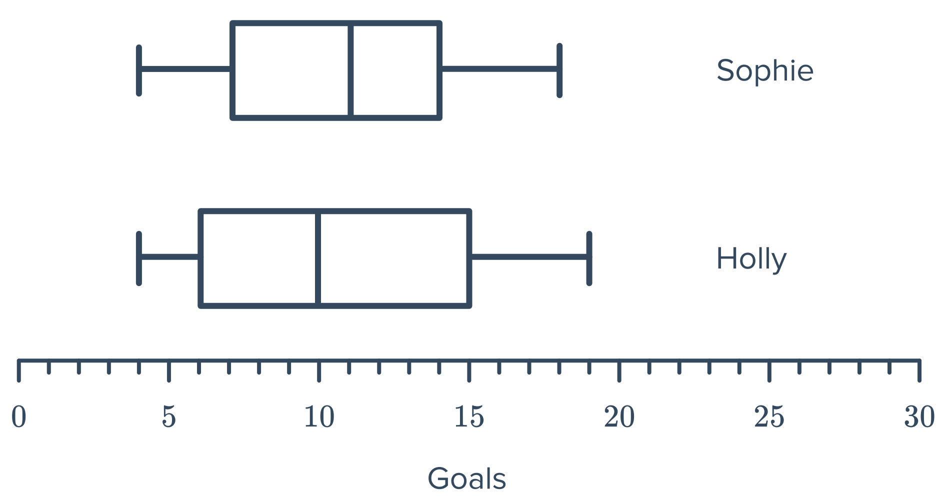


Class 9Q



b Which class tended to score better scores? Explain your answer.

22 The parallel box plots show the number of[redacted]s scored by two football players in each season:



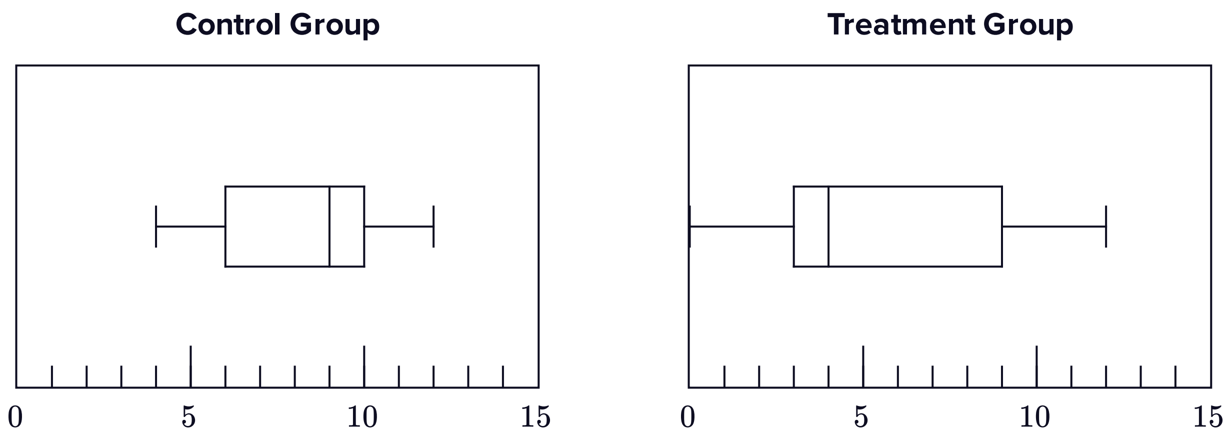
a Who has the greater median number of goals for the season?

b How many more goals did Holly score in her best season compared to Sophie in her best season?

c What is the difference between the median number of goals scored in a season by each player?

d What is the difference between the interquartile range for both players?

23 The boxplots summarize results from a medical study. The treatment group received an experimental drug to relieve cold symptoms, and the control group received a placebo. The boxplots show the number of days each group continued to report symptoms:



Answer true or false to the following:

a There is an outlier in the treatment group of 16.

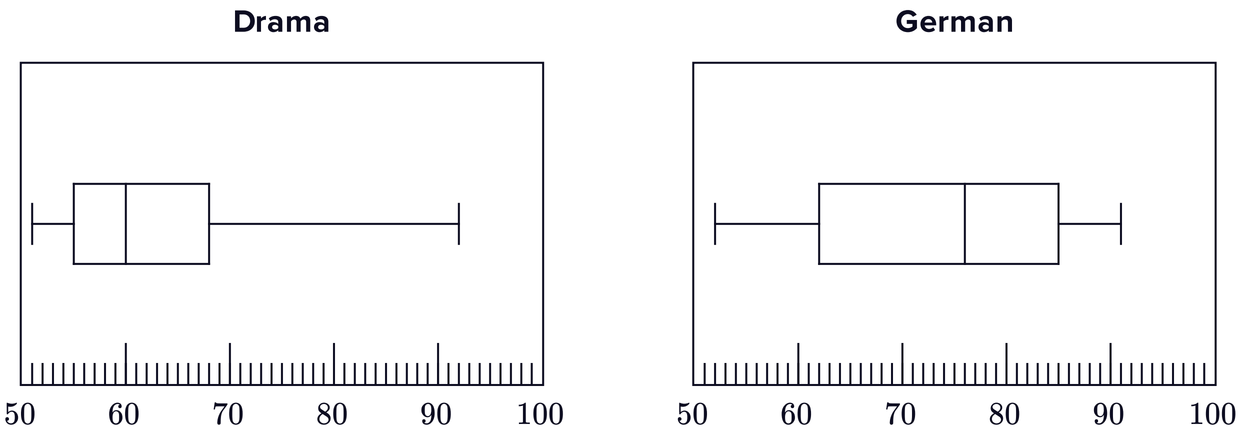
b The control group has the greater median number of days.

c The skew is more prominent in the treatment group.

d In the treatment group, cold symptoms lasted 0 to 12 days (range $= 12$) versus 4 to 12 days (range $= 8$) for the control group.

e It appears that the drug had a positive effect on patient recovery.

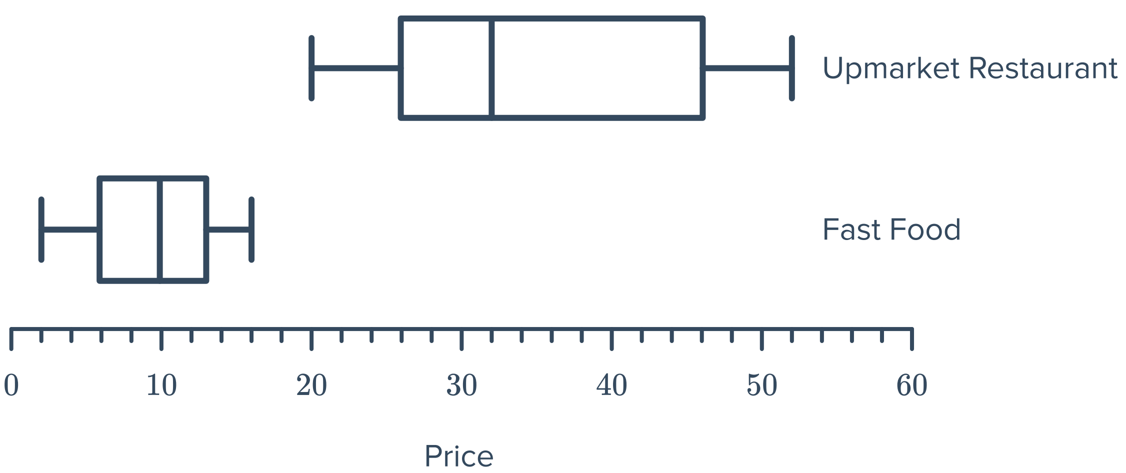
24 The test scores of 11 students in Drama and German are listed below:



a In which subject does there appear to be an outlier?

b In which subject did the students perform better?

25 The parallel box plots shows the prices, in dollars, of the items on the menu of an upmarket restaurant and the menu of a fast food restaurant:



a Which restaurant has the greater median price for the items they sell?

b What is the difference between the median prices of the items sold by each restaurant?

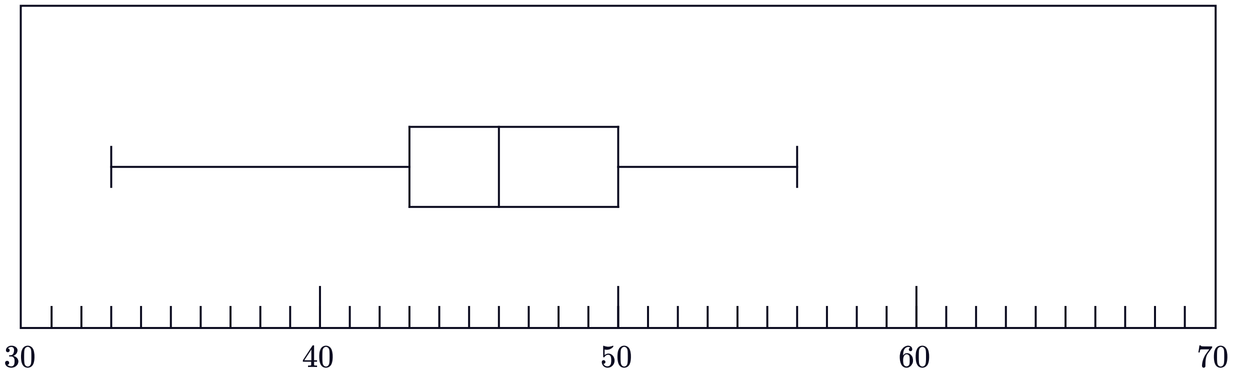
c Which restaurant has a greater price range for the items on the their menu?

d What is the price difference between the most expensive items sold by each restaurant?

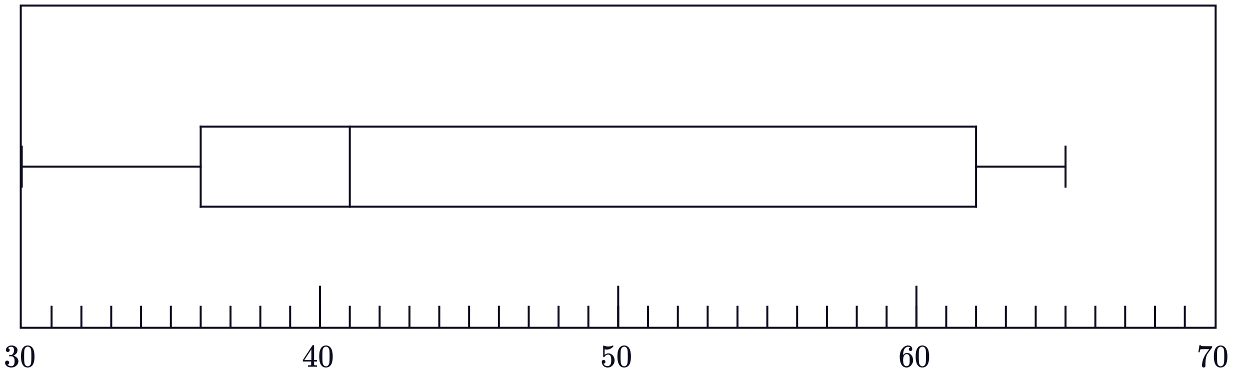
e What amount of the cheapest item at the fast food restaurant could be bought for the same price of the most expensive item at the upmarket restaurant?

26 The following box-and-whisker plots displ[illegible]tistics on the points scored by two basketball teams in each of their matches last season:

Team A



Team B



a State the median score for Team A.

b State the median score for Team B.

c Calculate the range for Team A's scores.

d Calculate the range of Team B's scores.

e Calculate the interquartile range of Team A's scores.

f Calculate the interquartile range of Team B's scores.

27 The data below represents how long each person in two different groups could hold their breath for, measured to the nearest second:

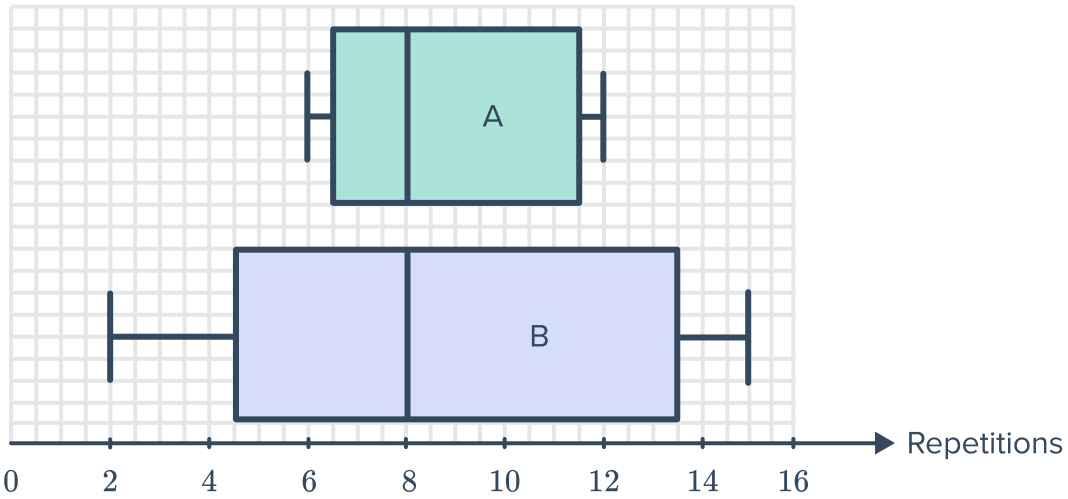
- Group A: $13, 17, 22, 28, 32, 34, 44, 49, 55, 56, 63, 64, 66, 51, 39$
- Group B: $13, 18, 23, 28, 32, 36, 42, 44, 53, 56, 64, 67, 13, 45, 47$

a Find the five number summary for the times of people in Group A.

b Find the five number summary for the times of people in Group B.

c Draw a parallel box plots for this data.

28 Two weightlifters both record their numbe[redacted]epetitions with a $70\ \mathrm{kg}$ bar over 30 days. The results are displayed in the box plots below:



a Which weightlifter has the more consistent results?

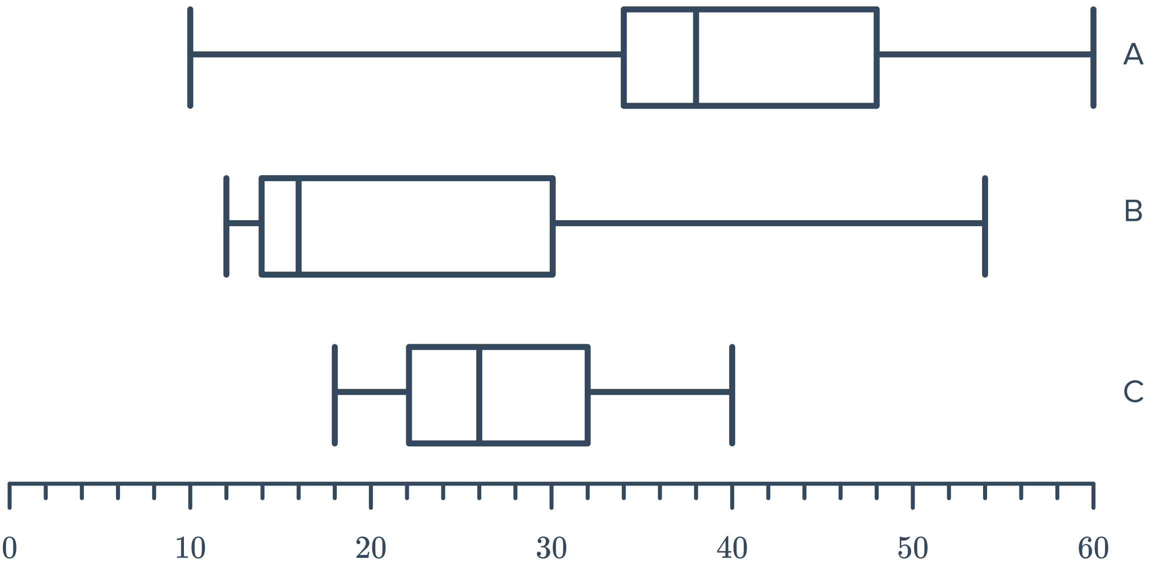
b Determine which of the following statistics supports your answer:

A The mean. **B** The range. **C** The mode. **D** The graph is positively skewed.

c Which statistic is the same for each weightlifter?

d Which weightlifter can do the most repetitions of $70\ \mathrm{kg}$?

29 A cinema is showing three films, labeled A, B and C. The ages of people watching each of the films are illustrated in the parallel box plots:



a Which film do you think has an adults only rating, restricting it to viewers 18 years of age and older? Explain your answer.

b Which film would you recommend for a group of 15 year olds to watch? Explain your answer.

c Which film would you recommend to a family of two parents in their 40's and two teenagers? Explain your answer.

Back-to-back stem and leaf plots

30 The test scores of 12 students in Music and French are listed below:

- Music: $79, 59, 74, 94, 51, 71, 93, 84, 69, 61, 86, 86$
- French: $62, 71, 64, 82, 83, 99, 87, 89, 66, 73, 59, 76$

Display the data as an ordered back-to-back stem and leaf plot.

31 The number of vehicles sold by two comp[illegible] each week from a dealership over three months was recorded in the back-to-back stem and leaf plot:

a Find the five number summary for the weekly number of vehicles sold over these three months by company A.

b Find the five number summary for the weekly number of vehicles sold over these three months by company B.

c Draw parallel box plots for this data.

Company A		Company B
5 0	**0**	3 9
8 7 4 1 1 0	**1**	0 2 2 2 3 7
9 2 1 0	**2**	0 1 1 7
9	**3**	1

Key: $2|1|0 = 12$ and 10

32 The back-to-back stem and leaf plot shows the number of pieces of paper used over several days by two classes, A and B:

a Find the five number summary for the number of pieces of paper used in class A.

b Find the five number summary for the number of pieces of paper used in class B.

c Draw parallel box plots for this data.

Class A		Class B
7	**0**	7
3	**1**	1 2 3
8	**2**	8
4 3	**3**	2 3 4
7 6 5	**4**	9
3 2	**5**	2

Key: $2|1|0 = 12$ and 10

33 The data below shows the results of a survey conducted on the price of concert tickets locally and the price of the same concerts at an international venue:

a What was the most expensive ticket price at the international venue?

b What was the median ticket price at the international venue?

c What percentage of local ticket prices were cheaper than the international median?

d At the international venue, what percentage of tickets cost between $\$90$ and $\$110$ (inclusive)?

e At the local venue, what percentage of tickets cost between $\$90$ and $\$100$ (inclusive)?

Local		International
7 5 2 2	**6**	0 5
9 6 5 4 0	**7**	2 3 8 8
9 6 5 3 0	**8**	2 3 7 8
8 7 4 3 1	**9**	0 1 6 7 9
5	**10**	0 2 3 5 8

Key: $6|1|2 = \$16$ and $\$12$

34 The stem and leaf plot below shows the p[illegible]f theatre tickets locally and internationally:

a Calculate the five number summary of the price of concert tickets at local venues.

b Calculate the five number summary of the price of concert tickets at international venues.

c Draw a parallel box plot for this data.

Local price		International price
7 6 3 0	**6**	1 8
8 6 4 3 2	**7**	3 5 5 9
9 6 5 1 1	**8**	1 5 7 9
8 7 5 2 0	**9**	1 3 4 6 8
1	**10**	1 2 4 7 8

Key: $2|6|0 = 62 \text{ and } 60$

35 The batting scores of two cricket teams, A and B, are recorded in the back-to-back stem and leaf plot below:

a Find the five number summary for the batting scores of team A.

b Find the five number summary for the batting scores of team B.

c Draw parallel box plots for this data.

Team A		Team B
9 5	**3**	2 3 6 6 8
8 8 5 5 4 1	**4**	2 9
9 5	**5**	0 8
	6	2

Key: $2|3|0 = 32 \text{ and } 30$

36 Ten participants had their pulse measured in beats per minute before and after exercise with results shown in the back-to-back stem and leaf plot on the right.

a Find the five number summary for the participants' pulse before exercise.

b Find the five number summary for the participants' pulse after exercise.

Pulse before		Pulse after
5 5 0	**5**	
9 9 7 4	**6**	
4 3	**7**	
0	**8**	4
	9	5 7 8
	10	3
	11	3 5 5
	12	0 1

Key: $2|8|0 = 82 \text{ and } 80$

37 The stem and leaf plot shows the number of books read in a year by a random sample of university and high school students:

University students		High school students
7	**0**	
6 6 3	**1**	0 0 3 5
4 3 2 1	**2**	1 2 4 4 6
9 8 8 6	**3**	1 8 9
8 2	**4**	0 1
	5	
	6	
3	**7**	

Key: $2|6|0 = 62 \text{ and } 60$

a Find the least number of books read by the university students.

b Calculate the median number of books for the high school students.

c Calculate the mean number of books for the university students. Round your answer to the nearest book.

38 The back-to-back stem and leaf plot shows the length (in minutes) of a random sample of phone calls made by Sharon and Tricia:

a Which girl made a 14 minute phone call?

b State the length of the longest phone call.

c Calculate the mean of Tricia's phone calls. Round your answer to the nearest minute.

d Calculate the median of Sharon's phone calls. Round your answer to the nearest minute.

Sharon's calls		Tricia's calls
3	**1**	3 4
7 6 4 3 2	**2**	6 7 8
9 8	**3**	2 4
4 3	**4**	1 2
7 6	**5**	6 7 8

Key: $2|6|0 = 62 \text{ and } 60$

39 The weight (in kilograms) of a group of men and women were recorded and presented in the stem and leaf plot below:

Weight of men		Weight of women
	5	0 1 2 3 4 4 4 5 5 5 7
9 8 8 7 6 6 6 5 3	**6**	0 2 2 3 4 7 7 8
6 4 3 2 2 1 0 0 0 0	**7**	0
0	**8**	

Key: $2|6|0 = 62 \text{ and } 60$

a Calculate the mean weight of the group of men. Round your answer to one decimal place.

b Calculate the mean weight of the group of women. Round your answer to one decimal place.

c Which group is heavier, on average?

40 The Cancer Council surveyed 60 random [redacted], asking them approximately how many hours they spent in the sun in the last month. The responders were split up into two groups, tourists and local residents and the results are shown below:

Tourists		Locals
9 8 7 7 6 5 4 4 3 1 1	**1**	2 4 4 5 5 5 6 8 9
9 5 1 0	**2**	2 3 5 9
9 7 6 1	**3**	0 2 5
7 6 6 5 4 3 1	**4**	0 0 2 3 6 6 7 8
9 6 3 0	**5**	2 2 3 5 8 9

Key: $6|1|2 = 12$ and 16

a What is the median number of hours that each group spent in the sun?

b If the two groups were combined, what would be the median number of hours spent in the sun?

c If the two groups were combined, what would be the range of responses?

41 The stem and leaf plot shows the test scores of classes A and B:

a Find the greatest score in Class A.

b Find the greatest score in Class B.

c Calculate the mean score of Class A. Round your answer to one decimal place.

d Find the mean score of Class B. Round your answer to one decimal place.

e Calculate the combined mean of the two classes correct to two decimal places.

Class A		Class B
6 5 3	**6**	3 5
6 3 2 0	**7**	2 5 7 7
4 2 0	**8**	1 3 3 6
	9	2 4

Key: $2|6|0 = 62$ and 60

42 The stem and leaf plot shows the number sserts ordered at two hotels over several randomly chosen days of a month:

a Which hotel had the largest range in number of desserts ordered?

b Which hotel had the greatest median number of desserts ordered?

Hotel A		Hotel B
3	**0**	
4 3 2	**1**	3 4
7 6	**2**	7
4 3	**3**	3 4
6	**4**	6 7
2	**5**	2 3 4

Key: $2|6|0 = 62$ and 60

Tables

43 Marge grows two different types of bean plants. She records the number of beans that she picks from each plant for 10 days in the table below:

Plant A	10	4	4	5	7	10	3	3	9	10
Plant B	8	7	5	5	9	7	8	7	5	6

a Calculate the mean number of beans picked per day for Plant A. Round your answer to one decimal place.

b Calculate the mean number of beans picked per day for Plant B. Round your answer to one decimal place.

c Find the range for Plant A.

d Find the range for Plant B.

e Which plant produces more beans on average? Explain your answer.

f Which plant has a more consistent yield of beans? Explain your answer.

44 The beaks of two groups of bird are measu[redacted]n mm, to determine whether they might be of the same species. Results are displayed in the table:

Group 1	33	39	31	27	22	37	30	24	24	28
Group 2	29	44	45	34	31	44	44	33	37	34

a Calculate the range for Group 1.

b Calculate the range for Group 2.

c Calculate the mean for Group 1. Round your answer to one decimal place.

d Calculate the mean for Group 2. Round your answer to one decimal place.

e Determine which of the following is the most appropriate statement to describe the set of data:

A Although the ranges are similar, the mean values are significantly different indicating that these two groups of birds are of the same species.

B Although the ranges are similar, the mean values are significantly different indicating that these two groups of birds are not of the same species.

C Although the mean values are similar, the ranges are significantly different indicating that these two groups of birds are not of the same species.

D Although the mean values are similar, the ranges are significantly different indicating that these two groups of birds are of the same species.

45 The number of goals scored by Team 1 and Team 2 in a football tournament are recorded in the following table:

a Find the total number of goals scored by both teams in Match C.

b Find the total number of goals scored by Team 1 across all the matches.

c Find the mean number of goals scored by Team 1.

d Find the mean number of goals scored by Team 2.

Match	Team 1	Team 2
A	2	5
B	4	2
C	5	1
D	3	5
E	2	3

46 The ages of employees at two competing [redacted]ood restaurants on a Saturday night are recorded. Some statistics are given in the following table:

a If the data for each restaurant was represented using a histogram, what would the likely shape of the histogram for Berger's Burgers be?

b Which restaurant likely has the oldest employee on the night the data is recorded?

c Which restaurant has the most consistent ages among employees? Explain your answer.

d Which restaurant has an older workforce? Explain your answer.

	Mean	Median	Range
Berger's Burgers	18	17	6
Fry's Fries	18	19	2

47 The table shows the scores of Student A and Student B in 5 separate tests:

Test	Student A	Student B
1	97	78
2	87	96
3	94	92
4	73	72
5	79	86

a Find the mean of the scores of Student A.

b Find the mean of the scores of Student B.

c Calculate the combined mean of the scores of the two students.

d Which student scored the greatest result in a test?

e State the least score over the two tests.

48 The hours of sleep per night for two peop[redacted]r a two week period are shown below:

Person 1	8	5	10	7	9	7	6	10	6	9	7	7	10	5
Person 2	8	8	8	7	7.5	8	7.5	7	7	7	7.5	7	7	7.5

a Copy and complete the table, giving your answers correct to one decimal place where appropriate:

	Person 1	Person 2
Mean		
Median		
Mode		
Range		

b Which person seems to be the least consistent in their sleep habits? Explain your answer.

c Which person had the most sleep over the 14 nights? Explain your answer.

49 The salaries of men and women working the same job at the same company are given below:

Men	80 000	80 000	75 000	80 000	75 000	70 000	80 000
Women	70 000	70 000	75 000	70 000	70 000	80 000	75 000

a Copy and complete the table, giving your answers correct to two decimal place where appropriate:

	Men	Women
Mean		
Median		
Mode		
Range		

b Which gender seems to be paid better? Justify your answer.

50 Two English classes, each with 15 student  10 question multiple choice test, each with four possible answers (only one of which is correct). Their class results, out of 10, are below:

Class 1	3	2	3	3	4	5	1	1	1	4	2	2	3	3	2
Class 2	8	9	9	8	8	6	8	10	6	8	8	9	6	9	9

a Copy and complete the following table for the two sets of data:

	Class 1	Class 2
Mean		
Median		
Mode		
Range		

b Which class was more likely to have studied for their test? Explain your answer.